

Corporate Failure Classification

Using Accounting Variables

A Comparison of Logistic Regression,
Decision Tree, and Random Forest

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GitHub repository: <https://github.com/Berke-Pehli/ml-finance-bankruptcy-analysis>

Machine Learning for Finance

University of Bonn

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Academic Year 2025/2026

1 Introduction

Corporate failure can create large losses for investors, lenders, employees, and other stakeholders. For this reason, it is useful to study whether financial information can help distinguish companies in difficulty from companies that remain active. Accounting variables may contain signals about liquidity, profitability, debt, and company size. Previous bankruptcy-prediction research has used accounting and financial variables to distinguish between healthy and distressed companies (Lombardo et al., 2022). However, identifying failed companies is difficult because failure is rare and financial conditions can differ strongly across companies.

This study examines corporate failure as a binary classification problem. Each observation represents one company in one year and is labelled as either active or failed. The analysis uses 18 accounting variables, labelled X_1 to X_{18} . The version used in this study contains 78,682 company-year observations and 8,971 unique company identifiers between 1999 and 2018. It is based on the American corporate bankruptcy dataset published by Lombardo et al. (2022). Failed observations account for only 6.6% of the full dataset. This strong class imbalance means that ordinary accuracy can give a misleading impression of model performance. For example, a model can achieve high accuracy by predicting almost every observation as active, while failing to identify the companies of greatest interest.

To study the classification problem, I compare three machine-learning models: Logistic Regression, Decision Tree, and Random Forest. I also include a majority-class baseline that always predicts the most common class. Logistic Regression provides a simple and interpretable linear model. Decision Tree allows nonlinear relationships and rule-based splits. Random Forest combines many trees and can reduce the instability of a single Decision Tree (Breiman, 2001; James et al., 2023). These models offer a clear comparison between a linear method, a single nonlinear model, and an ensemble model.

I split the data at the company level rather than at the individual-row level. This choice is important because the same company can appear in several years. A row-level split could place one year of a company in the training sample and another year of the same company in the test sample. This would create company-level information leakage and could make the final results too optimistic. In the company-level split, the test sample contains companies that are not used during model training.

Because failed observations are rare, I use balanced class weights for the learned models. I select the model settings using precision-recall area under the curve, or PR-AUC, on an internal validation sample. PR-AUC is suitable for this study because it focuses on the model’s performance for the rare failed class (Davis and Goadrich, 2006). After the model settings are fixed, I refit the models on the complete non-test training sample and evaluate them once on the untouched final test sample.

The research question is:

How well can Logistic Regression, Decision Tree, and Random Forest classify failed and active company-year observations for companies not seen during model training, using 18 accounting variables?

The study does not predict whether companies will fail after 2018 or within a specific future period. The dataset does not define a clear forecasting horizon. Therefore, the results should be interpreted as evidence about corporate failure classification rather than as a future bankruptcy forecast. The aim is to compare the models, examine their strengths and weaknesses, and identify which method provides the best balance between detecting failed observations and limiting false alarms.

2 Data and Descriptive Analysis

2.1 Dataset and Unit of Observation

The analysis uses the American corporate bankruptcy dataset introduced by Lombardo et al. (2022). The dataset contains accounting information for publicly listed companies in the United States.

The final sample includes 78,682 company-year observations and 8,971 unique anonymized company identifiers between 1999 and 2018. The data contain 18 accounting variables, a company identifier, a year variable, and the company status. The data-quality checks found no missing values and no duplicated rows.

Each row represents one company in one year. A company can therefore appear more than once if information is available for several years. This repeated structure is important for the modelling design. I keep the company identifier to create the training and test samples, while I keep the year variable for descriptive analysis. I do not use company or year as model predictors.

The observations cover the years from 1999 to 2018. However, I do not treat the final years as a separate forecasting period. The training and test samples may contain observations from the same calendar years, but they contain different companies. The evaluation therefore measures performance for unseen companies rather than performance in a later time period.

2.2 Target and Accounting Variables

The original company status has two categories: active and failed. For the analysis, I code active observations as 0 and failed observations as 1. I treat failed observations as the positive class throughout the paper. This means that precision, recall, and F1-score refer to the model’s ability to identify the failed class.

The model inputs consist of 18 variables labelled X_1 to X_{18} . These variables describe different parts of a company’s financial position and performance. They include measures related to assets, liabilities, debt, profitability, revenue, expenses, and market value. Examples are current assets, inventory, receivables, net income, EBIT, EBITDA, retained earnings, long-term debt, current liabilities, total liabilities, sales, and market value.

The variables are reported as financial levels rather than newly created financial ratios. This means that some variables may also reflect company size. For example, larger companies may naturally have higher assets, sales, liabilities, and expenses. This point is important when interpreting the results and will be discussed again in the Critical Reflection section.

The variables are measured on different scales. I therefore standardize the inputs used by Logistic Regression. The tree-based models do not require standardization because they divide observations through decision rules rather than through distances or coefficient magnitudes.

Table 1: Summary of the corporate failure classification dataset

Characteristic	Value
Company-year observations	78,682
Unique company identifiers	8,971
Sample period	1999–2018
Accounting variables	18
Active observations	73,462
Failed observations	5,220
Failed share	6.6%

Notes: Company and year are retained for company-level splitting and descriptive analysis, but they are not used as model predictors.

2.3 Class Distribution and Annual Failure Rate

The target variable is strongly imbalanced. Of the 78,682 company-year observations, 73,462 are active and 5,220 are failed. Failed observations represent 6.6% of the full sample. The first panel of Figure 1

shows the difference between the two classes.

The share of failed observations also changes over time. The second panel of Figure 1 shows that the observed failure rate increases from 7.2% in 1999 to a peak of 9.4% in 2003. It then generally declines and reaches approximately 1.3% in 2018.

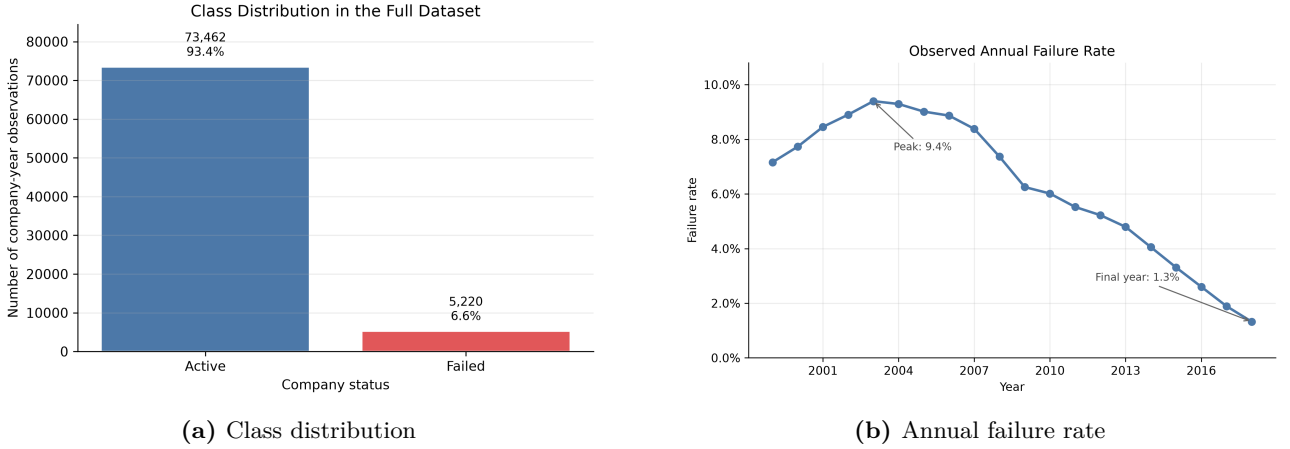


Figure 1: Target distribution and annual failure rate. Panel (a) shows the strong imbalance between active and failed observations. Panel (b) shows the descriptive annual failure rate. The annual pattern is descriptive and should not be interpreted as a causal effect or as evidence of model performance.

This imbalance creates an important evaluation problem. A model can obtain high accuracy by predicting almost every observation as active. Such a model would classify most observations correctly, but it would not identify the failed class. For this reason, I report accuracy only as supporting information. The main evaluation gives more attention to PR-AUC, failed-class precision, recall, and F1-score.

This pattern should be interpreted carefully. It may reflect changes in economic conditions, but it may also result from changes in the companies included in the dataset or in the way company status is recorded. I therefore use the figure only to describe the sample. It does not show that time caused the decline in the failure rate, and it is not evidence of model performance.

3 Methodology

3.1 Training, Validation, and Test Design

The modelling process uses separate training, validation, and test samples. I first divide the full dataset into an outer training sample and a final test sample. Approximately 80% of the companies are assigned to the training sample and 20% are assigned to the test sample. I use a fixed random seed of 42 so that the split can be reproduced.

The outer training sample contains 62,988 company-year observations from 7,176 company identifiers. The final test sample contains 15,694 observations from 1,795 company identifiers. The failed-observation shares are 6.4% in the training sample and 7.5% in the test sample.

I perform the split by company rather than by individual row. All yearly observations from one company remain in the same sample. This prevents one year of a company from appearing in the training data while another year of the same company appears in the test data. Otherwise, the model could partly learn company-specific information from the training sample. This would create company-level information leakage and could make the final performance appear better than it really is.

I divide the outer training sample again to create an internal model-training sample and a validation sample. I also perform this second split at the company level. The internal model-training sample

contains 50,323 observations from 5,740 company identifiers. The validation sample contains 12,665 observations from 1,436 different company identifiers. I use the validation sample to compare model settings, while the final test sample remains untouched.

This is not a chronological split. The training, validation, and test samples may contain observations from the same calendar years, but they contain different companies. The evaluation therefore measures generalization to unseen companies. It does not measure forecasting performance in a later time period.

3.2 Majority-Class Baseline

The first model is a majority-class baseline. It always predicts the most common class in the training data, which is the active class. The baseline does not learn a relationship between the accounting variables and the target. Its purpose is to provide a simple reference point for the machine-learning models.

Because active observations form more than 93% of the full dataset, this rule can achieve high accuracy. However, it does not identify any failed observations, so its recall for the failed class is zero. This shows why overall accuracy is not sufficient for evaluating models in an imbalanced classification problem. A useful corporate failure classifier should provide information beyond simply predicting that every company-year observation is active.

3.3 Logistic Regression

Logistic Regression is used as the main linear and interpretable model. It estimates the probability that an observation belongs to the failed class. The model links the accounting variables to the failure score through the logistic function:

$$P(Y = 1 | X) = \frac{1}{1 + \exp[-(\beta_0 + \beta_1 X_1 + \dots + \beta_{18} X_{18})]}. \quad (1)$$

Here, $Y = 1$ represents a failed observation. The coefficients $\beta_1, \dots, \beta_{18}$ describe the direction of the predictive relationship between each variable and the estimated failure score. A positive coefficient is associated with a higher predicted failure score, while a negative coefficient is associated with a lower score, holding the other variables fixed.

The accounting variables have very different numerical scales. For example, market value and total assets may be much larger than variables such as net income. I therefore standardize all predictors before fitting Logistic Regression. Standardization subtracts the training-sample mean and divides by the training-sample standard deviation. This gives each variable a mean close to zero and a standard deviation close to one within the training data.

The model uses standard L2 regularization with $C = 1.0$. L2 regularization adds a penalty when the model produces very large coefficients. This moves the coefficients closer to zero and can reduce overfitting. It is useful when several accounting variables contain related information because it can make the estimated coefficients more stable.

For example, current assets (X_1), market value (X_8), and total current liabilities (X_{14}) may all partly reflect the size and financial position of a company. If these variables contain overlapping information, Logistic Regression may otherwise assign an unnecessarily large or unstable coefficient to one of them.

L2 regularization reduces this problem by shrinking the coefficients while keeping all variables in the model. In scikit-learn, C controls the inverse of the regularization strength. A smaller value of C creates stronger regularization, while a larger value creates weaker regularization (James et al., 2023).

3.4 Decision Tree

A Decision Tree divides the observations into smaller groups by applying a sequence of decision rules. At each step, the model chooses a variable and a split point that improve the separation between active and failed observations. The final groups are called leaves, and each leaf produces a class prediction or failure score.

Decision Trees can capture nonlinear relationships and interactions without requiring them to be written into the model in advance. For example, the relationship between liabilities and failure may depend on the level of assets or profitability. A tree can represent this type of relationship through several connected splits.

A single Decision Tree can easily become too complex and fit noise in the training sample. I therefore compare a small number of settings during validation:

- Maximum tree depth: 3, 5, or 8
- Minimum observations per leaf: 20, 50, or 100

These values represent different levels of model complexity. A maximum depth of 3 produces a shallow and simple tree, while depths of 5 and 8 allow the model to capture more detailed relationships.

The minimum leaf sizes also control complexity. A value of 20 allows relatively small groups, while values of 50 and 100 require broader and more stable groups. This is important because failed observations are rare. A very small leaf could contain only a few failed observations and may therefore represent noise rather than a general pattern.

The grid is intentionally small. Its purpose is to compare simple, medium, and more flexible trees without creating an unnecessarily large tuning exercise. The specification with the highest validation PR-AUC uses a maximum depth of 5 and a minimum leaf size of 50.

A Decision Tree is easy to understand, but small changes in the data can produce a different tree. This instability is one reason for also using Random Forest.

3.5 Random Forest

Random Forest is an ensemble method that combines many Decision Trees. Each tree is fitted using a random sample of the training observations. In addition, only a random subset of variables is considered at each split. The final prediction is based on the combined results of all trees.

A single tree can have high variance because its structure may change strongly when the training data change. Random Forest reduces this problem by averaging many different trees. The individual trees may be unstable, but their combined prediction is usually more stable. Random Forest can also capture nonlinear relationships and interactions among the accounting variables (Breiman, 2001).

I compare a small number of settings during validation:

- Number of trees: 200
- Maximum depth: 5, 10, or no explicit limit
- Minimum observations per leaf: 10 or 30
- Variables considered at each split: the square root of the total number of predictors

These values represent different levels of model complexity. A maximum depth of 5 produces relatively simple trees, while a depth of 10 allows more detailed relationships. Using no explicit depth limit gives the trees more flexibility, although their growth is still restricted by the minimum leaf size.

Leaf sizes of 10 and 30 compare smaller, more detailed groups with larger and more stable groups. This is important because failed observations are rare, and very small leaves may be based on only a few failed cases.

The forest uses 200 trees because averaging a sufficiently large number of trees gives a more stable prediction, while keeping the computation manageable. At each split, the model considers the square root of the 18 predictors rather than all variables. This makes the individual trees less similar to each other, which improves the benefit of averaging them.

The grid is intentionally small and compares a few meaningful settings. The specification with the highest validation PR-AUC uses 200 trees, no explicit maximum depth, a minimum leaf size of 30, and the square-root rule for the number of variables considered at each split.

3.6 Class Imbalance and Classification Threshold

Failed observations form only a small part of the dataset. I therefore use balanced class weights for the learned models. This gives more importance to errors involving the rare failed class during model fitting. Without this adjustment, a model may focus mainly on the much larger active class.

Balanced class weights do not create new observations and do not change the test data. They only change how strongly errors from each class influence the fitting process. An error on a failed observation receives more weight than an error on an active observation because failed observations are much less common.

I use a classification threshold of 0.50 to convert each model score into a predicted class. If the predicted failure score is at least 0.50, I classify the observation as failed. If the score is below 0.50, I classify it as active.

The value 0.50 is a simple default rule, but it is not necessarily the best choice for a real financial application. If I lowered the threshold, the models would classify more observations as failed. This would usually increase recall because more actual failures would be detected, but it would also create more false alarms.

If I raised the threshold, the models would classify fewer observations as failed. This would usually improve precision, but some failed observations would be missed.

The best threshold therefore depends on the purpose of the model. For example, a bank that wants to avoid missing risky companies may prefer a lower threshold. A company with limited review capacity may prefer a higher threshold to reduce false alarms.

The dataset does not provide the financial costs of missed failures, false alarms, or manual reviews. I therefore do not optimize the threshold and use 0.50 as one consistent rule for comparing all models.

Balanced class weights also affect the estimated scores. Because I do not perform probability calibration, the scores should not be interpreted as exact real-world probabilities. For example, a score of 0.70 should not automatically be interpreted as a literal 70% probability of failure.

3.7 Model Selection and Evaluation Metrics

I select the Decision Tree and Random Forest settings using PR-AUC on the internal validation sample. I do not use the final test sample for model selection. PR-AUC summarizes the relationship between precision and recall across different possible score thresholds.

It is especially useful when the positive class is rare because it focuses directly on the model's performance for failed observations (Davis and Goadrich, 2006).

Precision measures how many observations classified as failed are actually failed:

$$\text{Precision} = \frac{TP}{TP + FP}. \quad (2)$$

Recall measures how many of the actual failed observations are identified by the model:

$$\text{Recall} = \frac{TP}{TP + FN}. \quad (3)$$

The F1-score combines precision and recall:

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (4)$$

Here, TP represents failed observations correctly classified as failed. FP represents active observations incorrectly classified as failed. FN represents failed observations incorrectly classified as active.

Accuracy measures the share of all observations classified correctly:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}. \quad (5)$$

TN represents active observations correctly classified as active.

I report accuracy as supporting information, but I do not use it as the main model-selection measure. The majority-class baseline shows that high accuracy can be achieved without identifying any failed observations.

I use PR-AUC as the main ranking and model-selection metric. I report failed-class precision, recall, and F1-score at the common classification threshold of 0.50. I also retain the confusion-matrix counts so that the numbers of correct classifications, missed failures, and false alarms remain visible.

3.8 Final Refitting and Test Evaluation

After I select the model settings with the validation sample, I combine the internal model-training and validation observations again. I then refit each learned model on the complete outer training sample of 62,988 observations. This allows the final models to use all available non-test information instead of leaving the validation observations unused.

The validation sample was previously used only to choose the Decision Tree and Random Forest settings. Once these choices are fixed, it can be included in the final training process. This does not create information leakage because the final test companies remain completely separate and are not used for model fitting or selection.

I then evaluate the refitted models once on the untouched final test sample. I do not change any model setting after seeing the test results. This is important because using test performance to choose or adjust a model would make the final results too optimistic.

The final comparison includes the majority-class baseline, Logistic Regression, Decision Tree, and Random Forest. I report accuracy, PR-AUC, failed-class precision, failed-class recall, failed-class F1-score, and the confusion-matrix counts.

4 Results

4.1 Validation Results and Model Selection

I first compared the model specifications on the internal validation sample. I used validation PR-AUC as the main selection measure because failed observations are rare. Precision-recall analysis is useful in this setting because it focuses on the model’s performance for the failed class and shows the trade-off between detecting failures and creating false alarms. I did not use accuracy for model selection because a model can achieve high accuracy by mainly predicting the active class (Davis and Goadrich, 2006).

Table 2 reports the validation PR-AUC values. Random Forest achieved the highest value of 0.151. Logistic Regression reached 0.147, while the selected Decision Tree reached 0.146. The majority-class baseline produced a PR-AUC of approximately 0.066. Its PR-AUC is 0.075 in the final test sample because the two samples have different failed-class shares.

Table 2: Validation PR-AUC used for model selection

Model	Validation PR-AUC
Majority-class baseline	0.066
Logistic Regression	0.147
Decision Tree	0.146
Random Forest	0.151

Notes: The majority-class baseline assigns the same score to every observation, so its PR-AUC is approximately equal to the failed-class share in the evaluation sample.

Based on this result, I selected Random Forest as the main model before evaluating the final test sample. The selected Decision Tree had a maximum depth of 5 and a minimum leaf size of 50. The selected Random Forest used 200 trees, no explicit maximum depth, a minimum leaf size of 30, and the square-root rule for the number of variables considered at each split.

The differences in validation PR-AUC are not large. However, Random Forest provided the strongest validation result among the models considered. The final test sample was not used to make this decision.

4.2 Final Test Performance

After fixing the model settings, I refitted the models on the complete outer training sample and evaluated them on the final test sample. Table 3 reports the final results at the common classification threshold of 0.50.

Table 3: Final test performance at the common classification threshold of 0.50

Model	Accuracy	PR-AUC	Failed precision	Failed recall	Failed F1
Majority-class baseline	0.925	0.075	0.000	0.000	0.000
Logistic Regression	0.361	0.151	0.095	0.879	0.171
Decision Tree	0.607	0.131	0.113	0.616	0.190
Random Forest	0.787	0.172	0.168	0.465	0.247

Notes: Failed observations are treated as the positive class. Accuracy, precision, recall, and F1-score are calculated at the threshold of 0.50. PR-AUC summarizes ranking performance across possible classification thresholds.

The majority-class baseline achieves an accuracy of 0.925 because most test observations are active. However, it classifies every observation as active. Its failed-class precision, recall, and F1-score are therefore zero. This confirms that accuracy alone gives a misleading picture in this dataset.

Logistic Regression reaches the highest failed-class recall of 0.879. This means that it identifies 87.9% of the failed observations. However, its precision is only 0.095. In other words, only 9.5% of the observations that it classifies as failed are actually failed. The model therefore detects most failures, but it also produces many false alarms.

The Decision Tree provides an intermediate result. Its recall of 0.616 is lower than the recall of Logistic Regression, but its precision of 0.113 is slightly higher. Its PR-AUC of 0.131 is the lowest among the three learned models on the final test sample.

Random Forest provides the strongest overall balance. It achieves the highest final-test PR-AUC of 0.172, the highest failed-class precision of 0.168, and the highest failed-class F1-score of 0.247. Its recall

is 0.465, so it identifies fewer failed observations than Logistic Regression and Decision Tree. However, its failure flags are more reliable, and it produces a better balance between precision and recall.

These results support the validation decision. Random Forest was selected before the final test evaluation, and it also produces the strongest overall performance on the unseen test companies.

4.3 Performance Trade-offs and Precision-Recall Curves

Figure 2 compares the final-test metrics across the four models. The figure shows that the choice of model depends on the evaluation objective. Logistic Regression gives priority to identifying failed observations, while Random Forest provides a stronger balance between detecting failures and limiting false alarms.

The accuracy panel should be interpreted carefully. The majority-class baseline has the highest accuracy because active observations dominate the test sample. However, it does not identify any failures. The failed-class metrics therefore provide a more useful view of model performance.

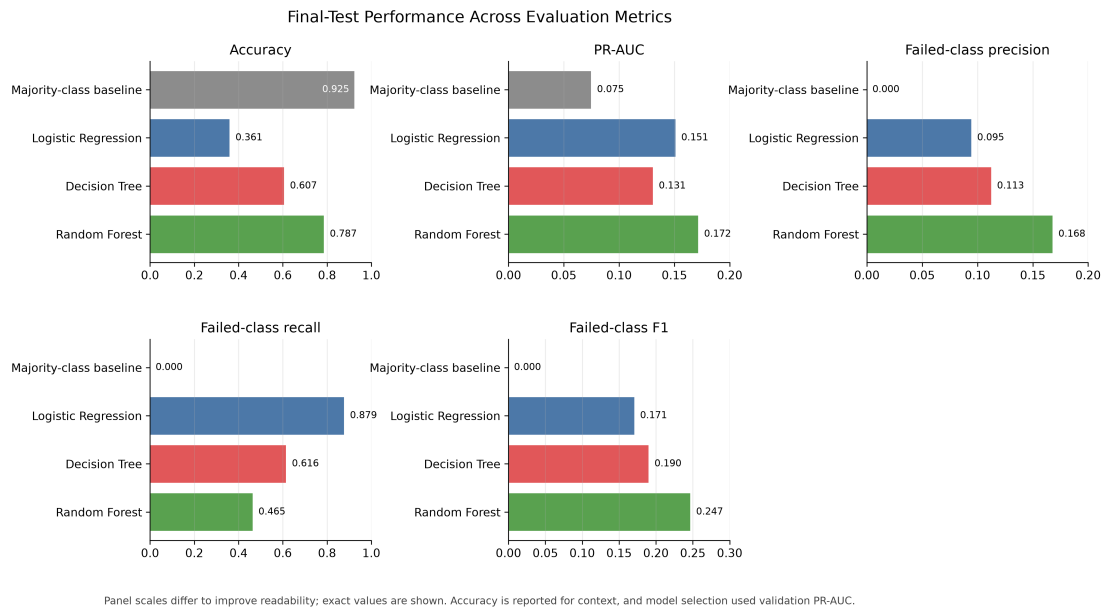


Figure 2: Final-test performance across evaluation metrics. Panel scales differ to improve readability, and exact values are shown. Accuracy is reported for context. Random Forest was selected using validation PR-AUC before final test evaluation.

Figure 3 shows the precision-recall curves for Logistic Regression, Decision Tree, and Random Forest. Each curve describes the relationship between precision and recall across many possible classification thresholds. The horizontal reference line represents the failed-observation share of approximately 7.5% in the final test sample.

A model with no useful ranking information would perform close to this reference level. All three learned models achieve PR-AUC values above the failed-class share, which shows that the accounting variables contain some useful information for ranking observations by failure risk.

Random Forest achieves the highest PR-AUC and generally provides the strongest precision-recall relationship. Logistic Regression reaches high recall, but its precision falls because many active observations also receive high failure scores. Decision Tree performs between the other models in some parts of the curve, but its overall PR-AUC is lower.

The precision-recall curves are not used to choose a new threshold. I use them to examine ranking performance across possible thresholds. The reported precision, recall, and F1-score still use the common threshold of 0.50.

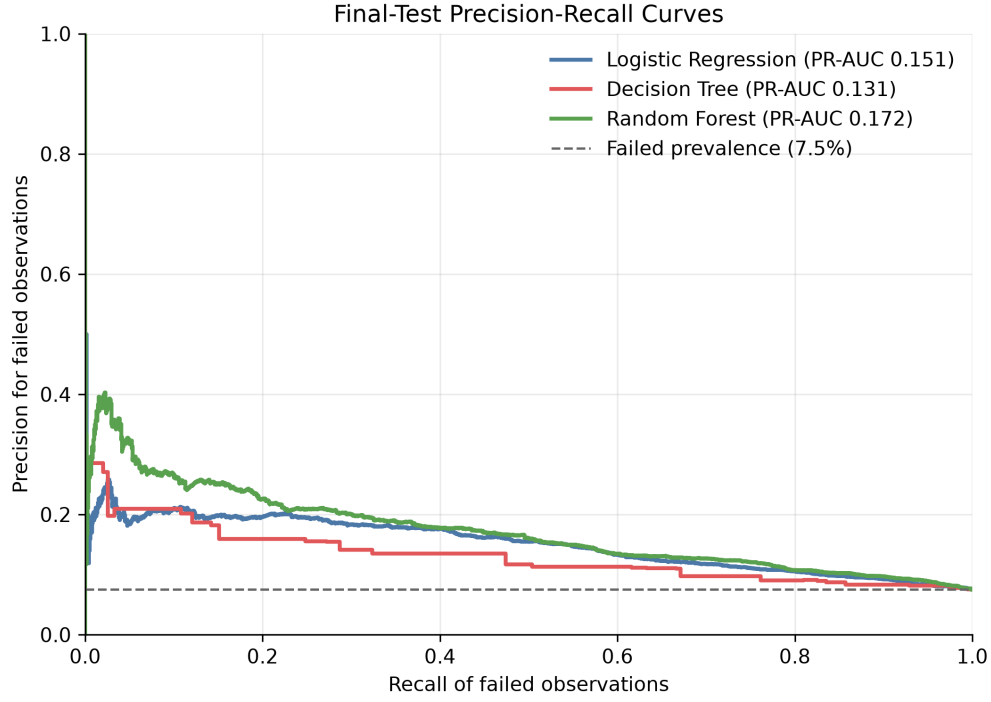


Figure 3: Final-test precision-recall curves. The curves compare the ranking performance of the three learned models across classification thresholds. The horizontal line shows the failed-class prevalence in the test sample.

Financial Meaning of the Classification Errors The two main error types have different financial meanings. A false negative occurs when a failed observation is classified as active. In a financial application, this could mean that a risky company is not identified. A false positive occurs when an active observation is classified as failed. This could create unnecessary manual reviews or overly cautious financial decisions.

Logistic Regression reduces false negatives but creates many false positives. Random Forest misses more failures, but its failure flags are more precise. Because the dataset does not provide the financial costs of these errors, I do not claim that one threshold is economically optimal.

5 Model Interpretation

5.1 Logistic Regression Coefficients

I first examine the coefficients from Logistic Regression because they show both the strength and the direction of the predictive relationship. The variables were standardized before fitting the model. Therefore, the coefficient sizes are more comparable even though the original accounting variables use different numerical scales.

A positive coefficient means that a higher value of the variable is associated with a higher predicted failure score, holding the other variables fixed. A negative coefficient means that a higher value is associated with a lower predicted failure score. These relationships are predictive rather than causal.

Panel (a) of Figure 4 shows the eight coefficients with the largest absolute values. Market value (X_8) has the largest negative coefficient, followed by current assets (X_1) and EBIT (X_{12}). In the fitted model, higher values of these variables are associated with a lower failure score. This result is financially reasonable because stronger market value, liquidity, and operating earnings may indicate a healthier financial position.

Total current liabilities (X_{14}) has one of the largest positive coefficients. This means that higher current liabilities are associated with a higher predicted failure score when the other variables are held fixed.

A possible financial explanation is that large short-term obligations may increase liquidity pressure. Depreciation and amortization (X_3) also has a positive coefficient. However, this variable should be interpreted carefully. Depreciation may reflect the size and asset structure of a company, and it may also be related to other accounting variables in the model. I therefore do not interpret it as a direct cause of failure.

The coefficient signs are useful because they show the direction of the model's relationship with the target. However, the magnitude of a coefficient can change when accounting variables contain overlapping information. L2 regularization also shrinks the coefficients toward zero. For these reasons, the coefficient ranking should be viewed as a description of the fitted Logistic Regression model rather than as a fixed economic ranking.

5.2 Random Forest Feature Importance

I also examine the feature importance from Random Forest because it is the strongest model in the final comparison. Random Forest importance measures how much the trees use each variable when splitting the observations. A variable receives a larger importance value when it repeatedly helps the forest separate active and failed observations.

Panel (b) of Figure 4 shows the eight variables with the largest importance values. Market value (X_8) has the highest importance. Net income (X_6), depreciation and amortization (X_3), current assets (X_1), and total long-term debt (X_{11}) also appear among the most important variables.

These variables cover several parts of a company's financial position. Market value reflects the market's overall assessment of the company. Net income and depreciation are related to financial performance. Current assets provide information about short-term resources, while long-term debt describes part of the company's financing structure. The result suggests that the forest uses information about company size, profitability, liquidity, and debt when separating the two classes.

Random Forest importance does not show whether a higher value increases or decreases the failure score. It only shows how strongly the model used the variable. For example, market value is highly important in the forest, but the importance value alone does not show the negative direction seen in Logistic Regression.

The importance values should also be interpreted carefully when variables are correlated. If two accounting variables contain similar information, the forest may divide the importance between them. The ranking therefore describes how this fitted Random Forest uses the variables. It does not prove that one variable is independently more important in an economic or causal sense (Breiman, 2001).

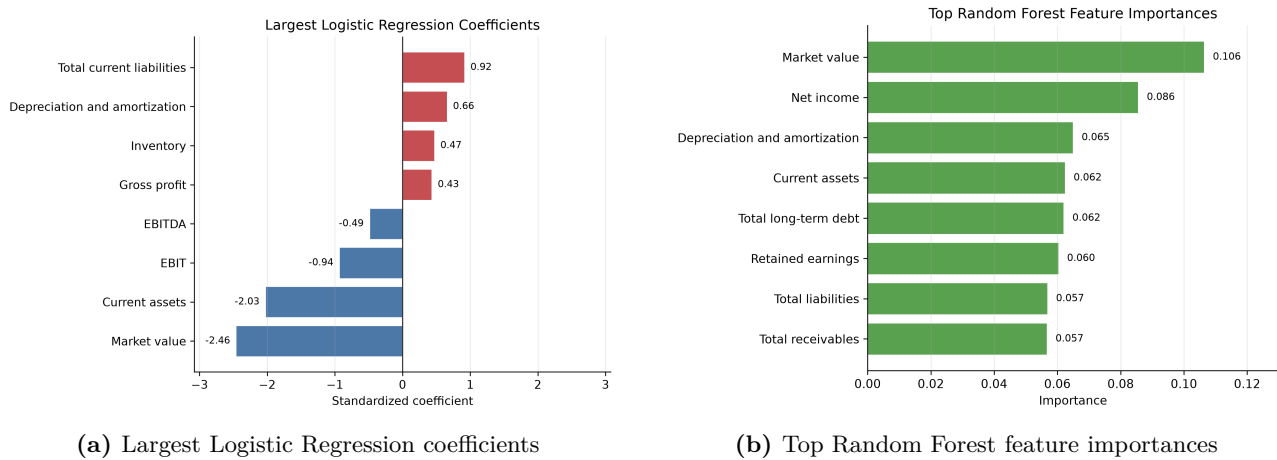


Figure 4: Interpretation of the fitted models. Panel (a) shows the largest standardized Logistic Regression coefficients and their directions. Panel (b) shows the variables most strongly used by Random Forest for splitting. Both figures describe predictive relationships rather than causal effects.

5.3 Comparing the Two Interpretation Methods

The two interpretation methods provide different information. Logistic Regression coefficients show both direction and relative strength. This makes the model easier to explain. However, Logistic Regression assumes a linear relationship in the log-odds and may not fully capture complex interactions.

Random Forest can capture nonlinear relationships and interactions, but its feature importance is less direct. It shows which variables the model used most, but it does not explain whether high or low values are associated with failure.

Some variables are important in both models. Market value (X_8), current assets (X_1), and depreciation and amortization (X_3) appear near the top of both rankings. This gives some evidence that these variables contain useful predictive information across different modelling approaches.

However, I do not interpret this agreement as evidence of causality. The project is designed to classify failed and active observations, not to estimate the causal effect of an accounting variable on corporate failure. The results may also be influenced by company size, correlations among the predictors, and the use of raw financial levels.

6 Critical Reflection

6.1 Limitations

This study has several limitations. First, the dataset does not define a clear future prediction horizon. I classify company-year observations using the failure status provided in the dataset, but I do not predict whether a company will fail in the following year or after 2018. The results should therefore be interpreted as corporate failure classification rather than as a future bankruptcy forecast.

Second, I use a company-level split rather than a chronological split. This prevents company-level information leakage because all observations from one company remain in the same sample. However, the training and test samples may contain observations from the same calendar years. The final evaluation therefore measures performance on unseen companies, not performance in a later economic period.

Third, the results are based on one fixed company split. The fixed random seed makes the analysis reproducible, but the reported results may change if different companies are assigned to the training and test samples. Repeated company-level splits or grouped cross-validation could provide more information about the stability of the model performance.

Fourth, I use the 18 accounting variables in their original financial levels. Variables such as assets, liabilities, sales, expenses, and market value may partly represent company size. The models may therefore use both financial condition and differences in company scale. Financial ratios or size-adjusted variables could provide a clearer view of liquidity, profitability, and leverage.

Fifth, I use balanced class weights because failed observations are rare. This helps the models pay more attention to the failed class, but it also affects their estimated scores. I do not perform probability calibration. The model outputs should therefore be interpreted as classification and ranking scores rather than as exact real-world probabilities of failure.

Sixth, I use a common classification threshold of 0.50. I do not optimize the threshold because the dataset does not provide the financial costs of missed failures, false alarms, or manual reviews. In a real application, the preferred threshold would depend on the purpose of the model and the decision-maker's tolerance for each type of error.

Finally, the interpretation results show predictive relationships rather than causal effects. A variable may help the model distinguish failed observations without directly causing corporate failure. The coefficient and feature-importance results may also be influenced by company size and overlapping information among the accounting variables.

6.2 Failures and Difficulties

The analysis did not produce a highly accurate corporate failure classifier. Although the learned models contain more useful information than the majority-class baseline, their performance remains limited. This is an important result because a financial machine-learning project does not automatically produce a strong or practically usable model.

The majority-class baseline shows the first difficulty. It achieves an accuracy of 0.925 by predicting every observation as active. However, it identifies no failed observations. Its failed-class recall and F1-score are both zero. This result shows that high overall accuracy can hide a complete failure to identify the class of financial interest.

Logistic Regression has a different problem. It identifies 87.9% of the failed observations, which gives it the highest recall. However, its failed-class precision is only 9.5%. This means that most observations classified as failed are actually active. The model reduces missed failures, but it creates a very large number of false alarms. Such a model could require more manual reviews than a financial institution can realistically perform.

The Decision Tree does not solve this trade-off. It reduces some false alarms compared with Logistic Regression, but its final-test PR-AUC is only 0.131. This is the lowest PR-AUC among the three learned models. The flexibility of the tree therefore does not produce a clear improvement in ranking failed observations.

Random Forest gives the strongest overall result, but its performance is still modest. Its final-test PR-AUC is 0.172 and its failed-class F1-score is 0.247. At the 0.50 threshold, it identifies only 46.5% of the failed observations. Therefore, more than half of the failed observations are missed. Its precision of 16.8% is better than the other learned models, but most failure flags are still false positives.

The validation results also show only small differences between the learned models. Random Forest has the highest validation PR-AUC, but the improvement over Logistic Regression and Decision Tree is limited. This means that the conclusion should remain careful. Random Forest is the strongest model among the methods I considered, but the evidence does not support the claim that it provides a complete or reliable corporate failure detection system.

The main difficulty is therefore the trade-off between recall and precision. Logistic Regression detects most failures but creates many false alarms. Random Forest gives more reliable failure flags but misses more failed observations. Without information about the financial cost of these errors, I cannot decide which trade-off would be best in practice.

6.3 What I Would Do Differently

If I repeated the project, I would first improve the definition of the target variable. I would create a clear future prediction horizon. For example, I would use accounting information from year t to predict whether a company fails in year $t + 1$. This would make the timing of the predictors and the failure event clear and would turn the analysis into a more realistic early-warning application.

I would then use a chronological evaluation design. The models would be trained on earlier years and evaluated on later years. A walk-forward approach would be especially useful because it would show whether the model continues to work when economic conditions and company populations change over time. I would still keep companies separated where necessary to avoid information leakage.

I would also improve the financial variables. Instead of using only the original accounting levels, I would construct ratios such as liquidity, leverage, profitability, and debt-service measures. I would also consider scaling financial levels by total assets or sales. These changes could reduce the influence of company size and produce variables that are easier to interpret financially.

I would evaluate the models across several company-level splits or use grouped cross-validation. This would show whether the model rankings and performance measures are stable or mainly caused by one particular split. I would report the average performance and its variation rather than relying only on

one result.

Finally, I would select the classification threshold using a financial cost framework. I would assign different costs to a missed failure, a false alarm, and a manual review. The preferred threshold could then be chosen according to the actual purpose of the model. A lender may place a high cost on missed failures, while an analyst with limited review capacity may place more importance on reducing false alarms.

I would focus on these improvements before adding many more machine-learning models. The main weaknesses of the current study come from the target definition, evaluation design, variables, and missing financial decision costs. A more complex model would not automatically solve these problems.

7 Conclusion

In this study, I examined how well Logistic Regression, Decision Tree, and Random Forest can classify failed and active company-year observations using 18 accounting variables. I evaluated the models on companies that were not used during model training. I also included a majority-class baseline to show whether the learned models provided information beyond simply predicting every observation as active. The majority-class baseline achieved an accuracy of 0.925, but it did not identify any failed observations. This result shows why accuracy alone is not suitable for this imbalanced classification problem. A model can classify most observations correctly while completely failing to detect the class of financial interest.

Logistic Regression identified the largest share of failed observations. Its failed-class recall was 0.879, which means that it detected 87.9% of the failures in the final test sample. However, its precision was only 0.095. Most of its failure predictions were therefore false alarms.

Random Forest provided the strongest overall balance among the models I compared. I selected it before the final test evaluation because it achieved the highest validation PR-AUC. On the final test sample, it also produced the highest PR-AUC of 0.172, the highest failed-class precision of 0.168, and the highest failed-class F1-score of 0.247. The Decision Tree gave an intermediate result, but it did not outperform Random Forest.

The model-interpretation results show that market value, current assets, profitability-related variables, and debt-related variables contain useful predictive information. Market value was especially important in both Logistic Regression and Random Forest. However, these relationships are predictive rather than causal, and some of the variables may also reflect company size.

I therefore conclude that the 18 accounting variables contain useful but limited information for distinguishing failed observations from active ones. Random Forest is the strongest model in this study when the goal is to balance precision and recall. Logistic Regression may be more suitable when identifying as many failed observations as possible is the main priority, although this comes with many false alarms.

The results do not support using any of the models as a complete financial decision system. A practical application would require a clearly defined future prediction horizon, chronological evaluation, improved financial variables, calibrated scores, and a classification threshold based on the financial costs of missed failures and false alarms. The main value of this study is therefore the careful comparison of the models and the explanation of their different strengths, weaknesses, and financial implications.

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